INCH-POUND

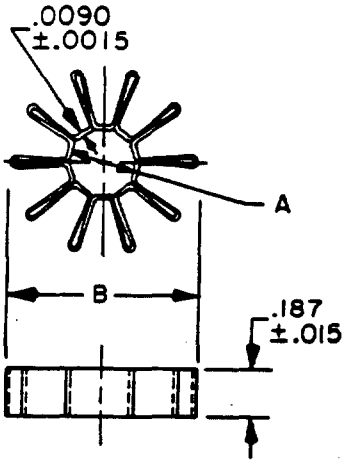
MIL-H-87111/3C
27 December 1989
SUPERSEDING
MIL-H-87111/3B
18 April 1986

MILITARY SPECIFICATION SHEET

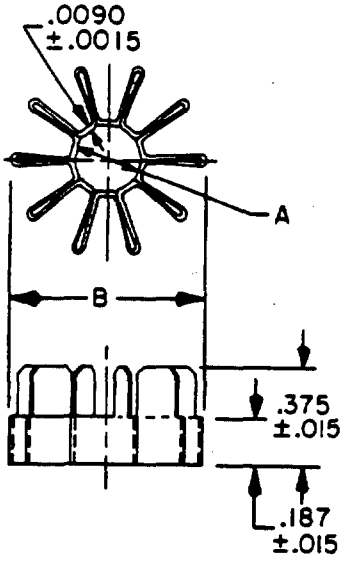
HEAT SINKS, ELECTRICAL-ELECTRONIC COMPONENT CLASS 1 AND CLASS 2 (PRESS-ON TYPE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-H-87111.



CONFIGURATION A



CONFIGURATION B

FIGURE 1. Heat sinks (press-on type).

© denotes changes

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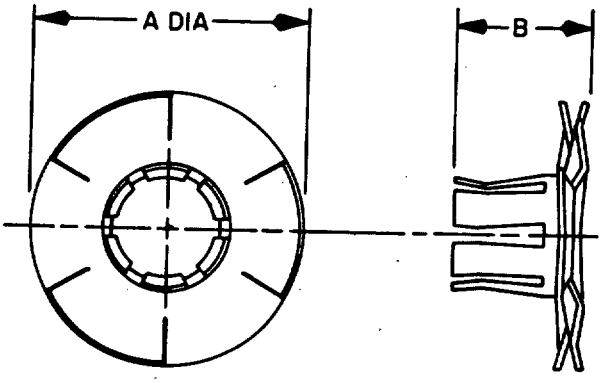
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Dash number 1/	JEDEC case size	Config-uration	Dimensions		Material
			A dia	B dia	
2-01	TO-18	A	.150 ±.010 (3.81 ±.25)	.640 ±.040 (16.26 ±1.02)	Beryllium copper (class 2)
2-02	TO-18	A	.150 ±.010 (3.81 ±.25)	.490 ±.040 (12.45 ±1.02)	
2-03	TO-5	A	.255 ±.010 (6.48 ±.25)	.550 ±.040 (13.97 ±1.02)	
2-04	TO-5	A	.255 ±.010 (6.48 ±.25)	.720 ±.040 (18.29 ±1.02)	
2-05	TO-18	B	.150 ±.010 (3.81 ±.25)	.640 ±.040 (16.26 ±1.02)	
2-06	TO-18	B	.165 ±.010 (4.19 ±.25)	.510 ±.040 (12.95 ±1.02)	
2-07	TO-5	B	.255 ±.010 (6.48 ±.25)	.720 ±.040 (18.29 ±1.02)	
2-08	TO-5	B	.280 ±.010 (7.11 ±.25)	.580 ±.040 (14.73 ±1.02)	
2-09	TO-8	B	.420 ±.010 (10.67 ±.25)	.830 ±.040 (21.08 ±1.02)	

1/ Complete dash number will include an additional letter to indicate finish (i.e., A, B, C, D, or H).

FIGURE 1. Heat sinks (press-on type) - Continued.

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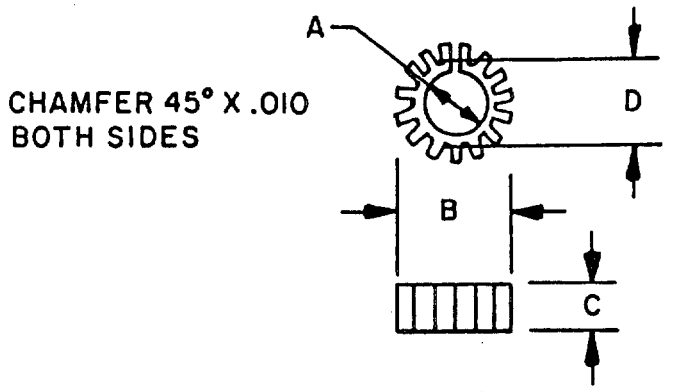
CONFIGURATION C

Dash number 1/	JEDEC case size	Dimensions		Material
		A dia	B dia	
2-10	TO-18	.500 ±.016 (12.70 ±.41)	.250 ±.016 (6.35 ±.41)	Beryllium copper (class 2)
2-11	TO-5	.750 ±.016 (19.05 ±.41)	.250 ±.016 (6.35 ±.41)	
2-12	TO-5	1.250 ±.016 (31.75 ±.41)	.360 ±.015 (9.14 ±.38)	
2-13	TO-8	1.250 ±.016 (31.75 ±.41)	.330 ±.015 (8.38 ±.38)	

1/ Complete dash number will include an additional letter to indicate finish (i.e., A, B, C, D, or H).

FIGURE 1. Heat sinks (press-on type) - Continued.

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CONFIGURATION D (see note 5)

Dash number 1/	JEDEC case size	Config-uration	Dimensions				Material
			A dia +.002 -.005	B dia ±.015	C ±.015	D dia ±.015	
1-14	TO-9	D	.294	.500	.250	.396	Aluminum alloy (class 1)
1-15	TO-9	D	.271	.500	.250	.396	
1-16	TO-5	D	.318	.500	.250	.396	
1-17	.255 dia	D	.245	.500	.250	.396	
1-18	TO-5	D	.314	.500	.250	.396	
1-19	TO-18	D	.177	.500	.250	.396	
1-20	TO-5	D	.281	.500	.250	.396	
1-21	TO-9	D	.228	.500	.250	.396	
1-22	.203 dia	D	.193	.500	.250	.396	
1-23	TO-58	D	.215	.625	.250	.440	
1-24	TO-42	D	.307	.625	.225	.440	
1-25	TO-42	D	.318	.625	.225	.440	
1-26	TO-18	D	.177	.625	.225	.440	
1-27	TO-5, TO-11	D	.330	.625	.375	.440	
1-28	TO-7	D	.353	.625	.375	.440	
1-29	.380 dia	D	.370	.625	.375	.440	
1-30	.407 dia	D	.397	.625	.375	.440	
1-31	TO-5, TO-11	D	.335	.625	.375	.440	
1-32	TO-5, TO-11	D	.322	.625	.375	.440	
1-33	TO-5, TO-11	D	.318	.625	.375	.440	
1-34	TO-9	D	.294	.500	.375	.396	
1-35	TO-9	D	.271	.500	.375	.396	
1-36	TO-5, TO-11	D	.318	.500	.375	.396	
1-37	.255 dia	D	.245	.500	.375	.396	
1-38	TO-5, TO-11	D	.314	.500	.187	.396	

1/ Complete dash number will include a letter to indicate finish (i.e., A, B, C, D, or H).

FIGURE 1. Heat sinks (press-on type) - Continued.

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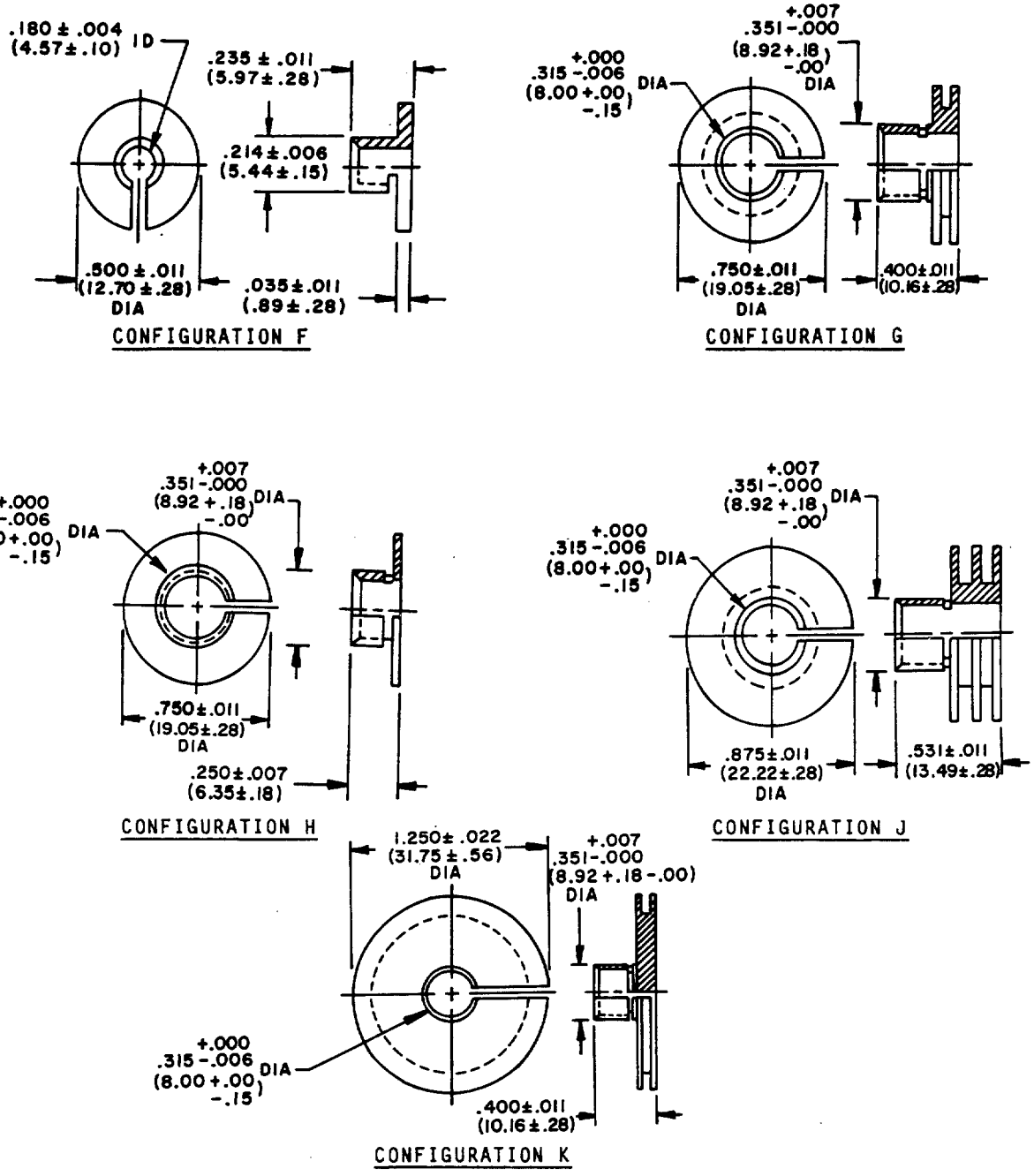
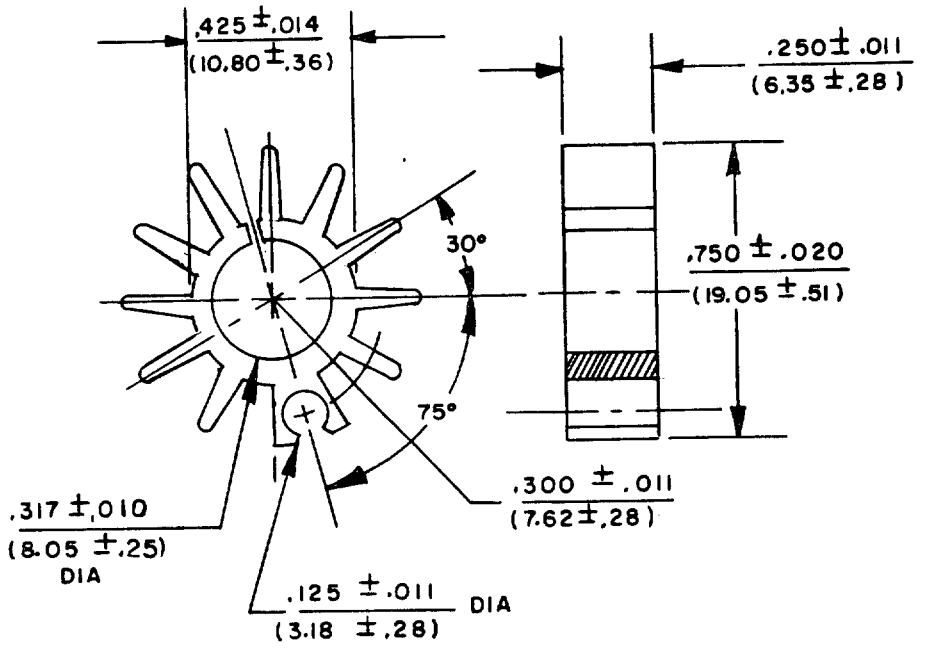
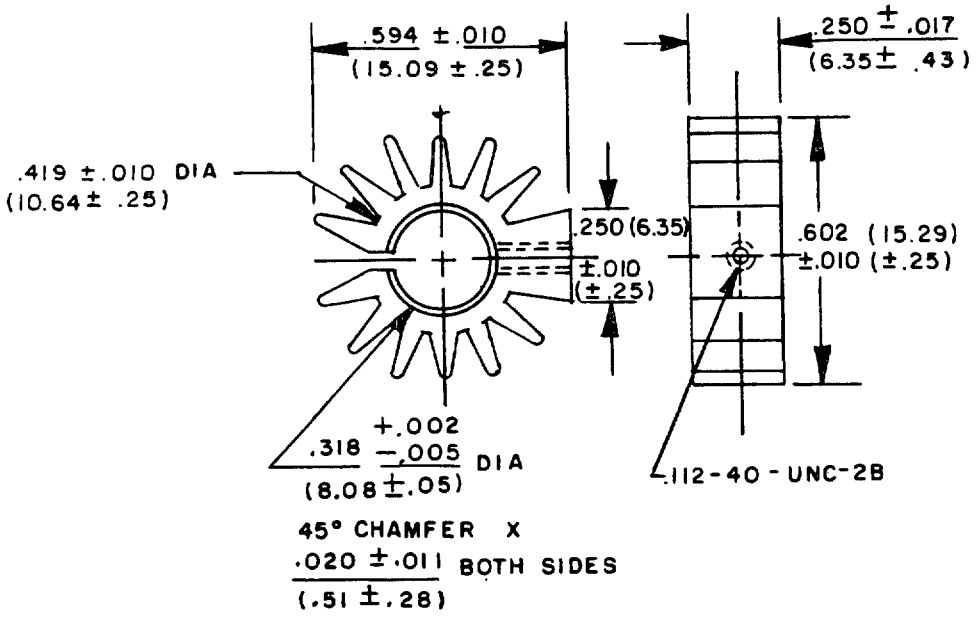


FIGURE 1. Heat sinks (press-on type) - Continued.

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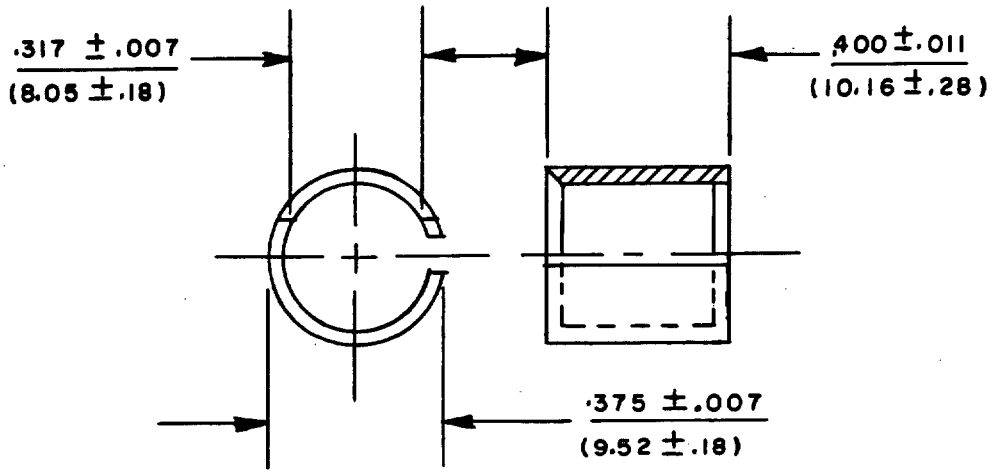
CONFIGURATION L



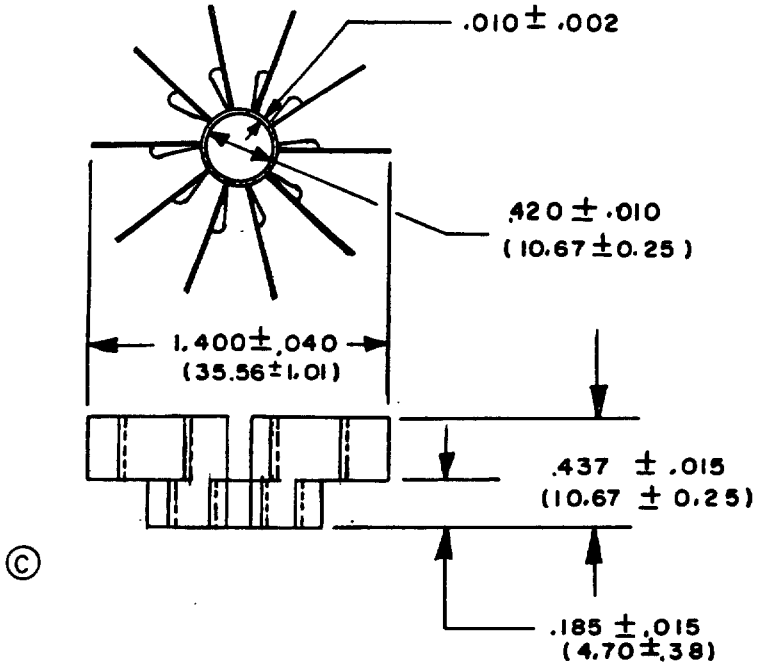
CONFIGURATION M

FIGURE 1. Heat sinks (press-on type) - Continued.

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CONFIGURATION N



CONFIGURATION P

FIGURE 1. Heat sinks (press-on type) - Continued.

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Dash number 1/	JEDEC case size	Configuration	Material
1-39	TO-18	F	Aluminum alloy
1-40	TO-5	G	
1-41	TO-5	H	
1-42	TO-5	J	
1-43	TO-5	K	
1-44	TO-5, 18, 92	L	
1-45	TO-5, 18, 92	M	
1-46	TO-5, 18, 92	N	
1-47	TO-8, 38	P	
2-48	TO-8, 38	P	Beryllium copper

1/ Complete dash number will include an additional letter to indicate finish (i.e., A, B, C, D, or H).

Inches	mm	Inches	mm
.112	2.84	.314	7.98
.177	4.50	.318	8.08
.187	4.75	.322	8.18
.193	4.90	.330	8.38
.215	5.46	.335	8.51
.225	5.72	.353	8.97
.228	5.79	.375	9.53
.245	6.22	.379	9.63
.250	6.35	.397	10.08
.271	6.88	.419	10.64
.281	7.14	.440	11.18
.294	7.47	.500	12.70
.307	7.80	.594	15.09
		.602	15.29

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.
4. A .010 inch (0.25 mm) radius is permissible on corners and edges.
5. The angle that the fins in configuration D make with respect to the centerline is optional (the shape of fins is optional).
6. Unless otherwise specified, tolerances are ±.005 (0.13 mm).

FIGURE 1. Heat sinks (press-on type) - Continued.

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REQUIREMENTS:

Dimensions and configurations: See figure 1.

Materials:

Aluminum alloy (class 1).

Beryllium copper (class 2).

Finish:

For beryllium copper; black chemical, insulating coating, or cadmium plating (finish B, C, or D, respectively).

For aluminum alloy; black anodize or hard anodize (finish A or H, respectively).

Thermal resistance: See table I.

TABLE I. Thermal resistance.

Dash number	°C/watt (unmounted)
2-01 through 2-13	80
1-14 through 1-23	90
1-24 through 1-26	93
1-27 through 1-33	65
1-34 through 1-38	76
1-39	92
1-40	46
1-41	54
1-42	35
1-43	21
1-44	48
1-45	37
1-46	68
1-47	28
2-48	28

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Part or Identifying Number (PIN): M87111/03- (plus dash number from figure 1. Second dash shall be replaced with a letter to denote finish.). Example: M87111/03-1H34.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - TD
Air Force - 11, 19, 99
DLA - ES

User activities:
Navy - AS, CG, MC, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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